

1

Electric charge

Distinguish charge types and explain conservation and quantisation.

p. 1

- Like charges repel; unlike charges attract.
- SI unit: coulomb (C).
- Conserved in an isolated system; quantised in multiples of e .

Self-check: What happens when two positive charges are brought near each other?

2

Coulomb's law

Use the inverse-square relation and identify force direction.

p. 2

- $F = k |q_1 q_2| / r^2$.
- Force acts along the line joining the charges.
- Net force is the vector sum of individual forces.

Self-check: What happens to force when separation doubles?

3

Electric field

Separate field from force and interpret field-line direction.

p. 3

- Vector quantity: magnitude and direction.
- SI unit: N/C.
- Field lines go away from positive charge and toward negative charge; they do not intersect.

Self-check: Why should a test charge be small?

4

Electric potential

Distinguish potential, potential difference and electric field.

p. 4

- SI unit: volt = joule/coulomb.
- Potential difference is change in potential energy per unit charge between two points.
- Zero potential does not necessarily mean zero electric field.

Self-check: Is electric potential a vector?